about point on circumference:

$$I_0 = \iint r^2 \, dA = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2a\cos\theta} r^2 \, r \, dr \, d\theta$$

Inner:

$$= \left[\frac{r^4}{4} \right]_0^{2a\cos\theta}$$

$$= 4a^4 \cos^4 \theta$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4a^4 \cos^4 \theta \, d\theta$$

$$= \frac{3\pi a^4}{2}$$

lec 15: change of variables
Changing variables in double integrals
Example: Area of ellipse with semi-axes a, b

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 \leq 1$$

$$\iint dx \, dy$$

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 \leq 1$$

Set

$$\frac{x}{a} = u, \qquad \frac{y}{b} = v$$

$$= \iint_{u^2+v^2 \leq 1} du \, dv$$

$$\frac{x}{a} = u \qquad \Rightarrow \qquad du = \frac{1}{a} dx$$

$$\frac{y}{b} = v \qquad \Rightarrow \qquad dv = \frac{1}{b} dy$$

$$du \, dv = \frac{1}{ab} dx \, dy$$

$$dx\,dy = ab\,du\,dv$$

$$= ab \iint_{u^2+v^2 \leq 1} du\,dv$$

$$= ab \times \text{Area(unit disk)}$$

$$= \pi ab$$

In general, find scaling factor

$$(dx\,dy \text{ vs. } du\,dv)$$

Example 2

$$u = 3x - 2y$$

$$v = x + y$$

to simplify

Relation between

$$dA = dx\,dy$$

and

$$dA' = du\,dv$$

Area scaling factor here does not depend on choice of rectangle.

$$A' = \begin{vmatrix} 3 & 1 \\ -2 & 1 \end{vmatrix} = 5$$

For any other rectangle, rectangle is also scaled by 5.
So,

$$dA' = 5\,dA$$

$$du\,dv = 5\,dx\,dy$$

$$dx\,dy = \frac{1}{5} du\,dv$$

$$\iint dx\,dy = \iint \frac{1}{5}\,du\,dv$$

General case

$$u = u(x, y)$$

$$v = v(x, y)$$

$$\Delta u = u_x \Delta x + u_y \Delta y$$

$$\Delta v = v_x \Delta x + v_y \Delta y$$

$$\begin{pmatrix} \Delta u \\ \Delta v \end{pmatrix} = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}$$

Small rectangle

Same argument

$$\langle \Delta x, 0 \rangle \longrightarrow \langle u_x \Delta x, v_x \Delta x \rangle$$

$$\langle 0, \Delta y \rangle \longrightarrow \langle u_y \Delta y, v_y \Delta y \rangle$$

$$\text{Area}' = \det(r)\, \Delta x\, \Delta y$$

$$\text{Jacobian } J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

Then

$$du dv = |J|\, dx dy = \left| \frac{\partial(u, v)}{\partial(x, y)} \right| dx dy$$

Ex 7: Polar Coordinates

$$x = r \cos\theta$$

$$y = r \sin\theta$$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta$$

$$= r$$

So

$$dxdy = |r|\, drd\theta$$

$$= r\, drd\theta$$

Remark

$$\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$$

So we compute the easiest one!

Ex 2

Compute

$$\int_0^1 \int_0^1 x^2 y\, dx\, dy$$

by changing variables to

$$u = x, \qquad v = xy$$

1) Area Element

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 0 \\ y & x \end{vmatrix} = x$$

so

$$dudv = x\, dxdy$$

$$dxdy = \frac{1}{x}\, dudv \qquad (x > 0)$$

2) Integral in terms of u, v

$$x^2y\,dxdy = x^2y \cdot \frac{1}{x}\,dudv$$

$$= xy\,dudv$$

Since

$$v = xy$$

we get

$$= v\,dudv$$

Hence,

$$\iint v\,dudv$$

3) Bounds

Switch to the uv-picture.

When

$$x = 1$$

then

$$u = 1$$

and

$$v = xy = y$$

When

$$y = 1$$

then

$$v = x = u$$

Also,

$$y = \frac{v}{u} = 1$$

Lecture 19: Vector Fields and Line Integrals

Vector Fields

$$\vec{F} = M\hat{\imath} + N\hat{\jmath}$$

where M and N are functions of x, y.
At each point (x, y), $\vec{F}$ is a vector that depends on (x, y).

Examples

Velocity in a fluid:

$$\vec{V}$$

Force field:

$$\vec{F}$$

Examples of Vector Fields

$$\vec{F} = 2\hat{\imath} + \hat{\jmath}$$

$$\vec{F} = x\hat{\imath}$$

$$\vec{F} = -y\hat{\imath} + x\hat{\jmath}$$

(Velocity field for uniform rotation at unit angular velocity)

Work and Line Integrals

$$W = (\text{force}) \cdot (\text{distance})$$

$$W = \vec{F} \cdot \Delta\vec{r}$$

Along a trajectory C, work adds up to

$$W = \int_{t_1}^{t_2} \vec{F} \cdot \frac{d\vec{r}}{dt}\, dt$$

$$= \lim_{\Delta r \to 0} \sum_i \vec{F} \cdot \Delta\vec{r} = \sum_i \vec{F} \cdot \left(\frac{\Delta\vec{r}}{\Delta t}\Delta t\right)$$

$$\frac{d\vec{r}}{dt}$$

Example:

$$\vec{F} = -y\hat{\imath} + x\hat{\jmath}$$

$$C_1: \quad x = t, \qquad y = t^2$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F} \cdot \frac{d\vec{r}}{dt}\, dt$$

$$\vec{F} = \langle -y, x \rangle = \langle -t^2, t \rangle$$

$$x = t, \qquad y = t^2$$

$$\frac{dx}{dt} = 1, \qquad \frac{dy}{dt} = 2t$$

$$= \int_0^1 (-t^2 + 2t^2)\, dt$$

$$= \int_0^1 t^2\, dt = \frac{1}{3}$$

Another way:

$$\vec{F} = \langle M, N \rangle \qquad d\vec{r} = \langle dx, dy \rangle$$

$$\vec{F} \cdot d\vec{r} = M dx + N dy$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C M dx + N dy$$

Method to evaluate:
Express x, y in terms of a single variable and substitute.

$$\int_C \vec{F} \cdot d\vec{r} = \int_C -y\, dx + x\, dy$$

In terms of t:

$$x = t, \qquad y = t^2$$

$$dx = dt, \qquad dy = 2t\, dt$$

$$= \int_C -t^2\, dt + t \cdot 2t\, dt$$

$$= \int_0^1 t^2\, dt = \frac{1}{3}$$

Note:

$$\int_C \vec{F} \cdot d\vec{r}$$

depends on the trajectory C but not on the parametrization.
For example, one could use

$$\begin{cases} x = \sin\theta \\ y = \sin^2\theta \end{cases} \qquad 0 \le \theta \le \frac{\pi}{2}$$

(not practical).

Geometric approach

$$d\vec{r} = \langle dx, dy \rangle = \hat{T}\, ds$$

$$\frac{d\vec{r}}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle = \hat{T}\frac{ds}{dt}$$

So,

$$\int_C \vec{F} \cdot d\vec{r} = \int_C M\, dx + N\, dy = \int_C \vec{F} \cdot \hat{T}\, ds$$

Example:
C = circle of radius a at the origin (counterclockwise)

$$\vec{F} = x\hat{i} + y\hat{j}$$

$$\vec{F} \perp \hat{T} \quad \text{so} \quad \vec{F} \cdot \hat{T} = 0$$

$$\int_C \vec{F} \cdot \hat{T}\, ds = 0$$

2) Same C

$$\vec{F} = -y\hat{i} + x\hat{j}$$

Now $\vec{F} \parallel \hat{T}$

$$\vec{F} \cdot \hat{T} = |\vec{F}| = a$$